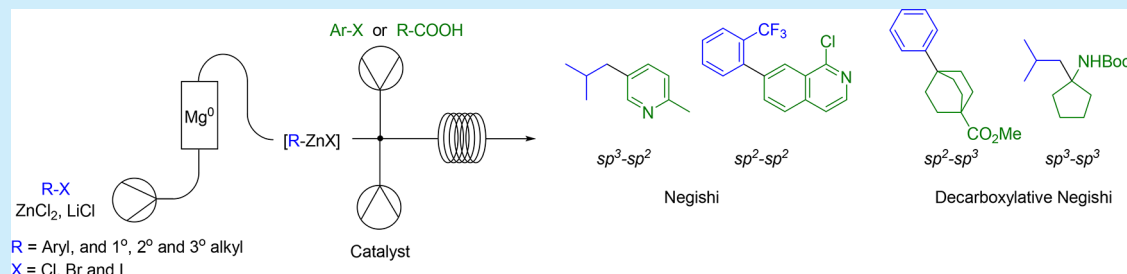


Generation and Cross-Coupling of Organozinc Reagents in Flow

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Supporting Information



ABSTRACT: A versatile flow synthesis method for in situ formation of organozinc reagents and subsequent cross-coupling with aryl halides and activated carboxylic acids is reported. Formation of organozinc reagents is achieved by pumping organic halides, in the presence of ZnCl_2 and LiCl , through an activated Mg^0 -packed column under flow conditions. This method provides efficient in situ formation of aryl, primary, secondary, and tertiary alkyl organozinc reagents, which are subsequently telescoped downstream to a Negishi or decarboxylative Negishi cross-coupling reaction. The described method offers access to a variety of C–C bond formations with organozinc reagents that are otherwise commercially unavailable or difficult to prepare under traditional batch reaction conditions.

Transition-metal-mediated C–C bond-forming reactions with organometallic reagents are widely used in organic synthesis.¹ Organoboranes are frequently employed cross-coupling partners in drug development due to their stability and abundant availability. In contrast, organozinc reagents are less commonly used due to their limited commercial availability and methods of preparation yet can provide a reactivity scope superior to that of organoboranes.² Organozinc reagents strike a balance in reactivity between organoboranes and Grignard or organolithium reagents. Negishi³ ($\text{sp}^3\text{--sp}^2$ and $\text{sp}^2\text{--sp}^2$) and decarboxylative Negishi⁴ ($\text{sp}^2\text{--sp}^3$ and $\text{sp}^3\text{--sp}^3$) cross-couplings are two attractive applications of organozinc reagents in medicinal chemistry settings. Formation of bench-stable organozinc reagents can be technically challenging due to their strong nucleophilic and basic character.⁵ Standard flask or batch synthesis of organozinc reagents is often complicated by inert and anhydrous atmosphere conditions and irregular Mg activation, factors that are easily remedied under flow chemistry conditions.

Flow chemistry technology is ideally suited for in situ synthesis of reactive intermediates with subsequent in-line trapping or reaction telescoping.⁶ Moreover, the potential to run moisture-sensitive reactions in a single, uninterrupted micro-reactor sequence is beneficial for rapid and efficient generation of large numbers of compounds with high purity and minimal byproducts.⁷ Modern flow chemistry instrumentation enables telescoping of multiple synthetic steps in one continuous flow and even allows for in-line aqueous workups and purification.^{6b,8}

Generation of organozinc reagents using flow chemistry by simply passing alkyl halides through an activated zinc packed

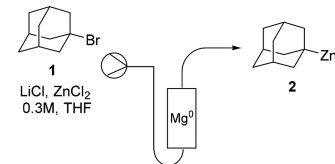
column was recently reported. These reagents were used in subsequent cross-coupling with aryl bromides.⁹ This method provides direct in situ access to organozinc reagents; however, it is limited to benzyl bromides and secondary alkyl iodides, warranting further investigation to expand substrate scope.

To satisfy our medicinal chemistry needs, we sought to identify a generalized flow protocol with broader substrate scope for the organozinc reagents that would enable telescoping in-line by cross-coupling with aryl halides and activated carboxylic acids.

Work by Knochel demonstrated an efficient batch method for the synthesis of organozinc reagents using Mg in the presence of LiCl and ZnCl_2 .¹⁰ As Mg is a stronger reducing agent than zinc, halogen/metal exchange¹¹ with Mg proceeds faster with higher substrate scope in generating organozincs.^{2b,12} Importantly, this method allows for the formation of sterically hindered organozincs such as adamantylzinc reagents. Based on this work, we investigated Mg -packed columns to form transient Grignard intermediates as precursors to organozinc reagents under flow conditions. Few groups have reported the synthesis of Grignard reagents using a Mg column in flow,¹³ yet direct transmetalation to the organozinc reagent is largely unexplored.

Our initial efforts focused on selecting the optimal Mg particle size and activation conditions for the Mg column by studying the conversion of 1-bromoadamantane (**1**) to its corresponding adamantylzinc reagent (**2**) (Table 1). The concentration of adamantylzinc was determined by quenching with iodine.¹⁴

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Table 1. Magnesium Activation and Particle Size Optimization with 1-Bromoadamantane (1) in Flow^a


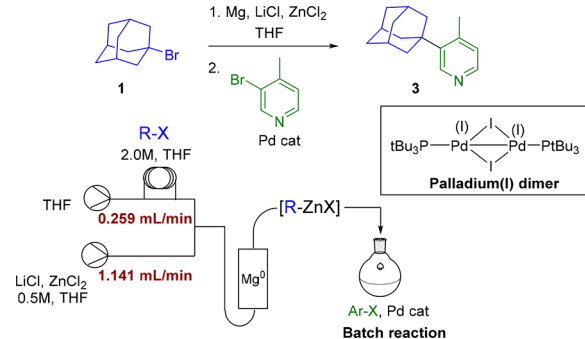
entry	Mg ⁰ particle size	temp (°C)	<i>r_t</i> (min) ^b	conversion (%) ^c
1	turnings	25	5	<5
2	turnings	50	5	10
3	turnings	100	10	75
4	−50 mesh ^d	25	10 ^e	75

^aReaction condition: **1** (0.3 M in THF), ZnCl₂ (1.1 equiv), LiCl (1.1 equiv). ^bResidence time (*r_t*) as measured in an Omnifit 150 mm × 6.6 mm column. ^cConversion was determined by quenching with iodine.¹⁴ ^dAs supplied by Aldrich, catalog# 253987. ^eOmnifit 150 mm × 15 mm column.

Reactions were first carried out using Mg turnings which were activated with TMSCl (2 M in THF) and 1,2-dibromoethane. Under these conditions, precipitation of Mg salts was observed at the end of the column and the exit tubing, which led to system clogging. Further optimization suggested that lowering the concentration of 1,2-dibromoethane (0.3 M in THF) was necessary to avoid precipitation of Mg salts during activation. Once the Mg column was activated, a combined solution of organohalide, LiCl, and commercially available ZnCl₂ in THF flowed through the column reactor at rt to generate the corresponding organozinc reagents. Even though Mg turning activation was successful (Table 1, entry 3), it required increased temperatures and longer residence times to achieve high transmetalation conversion to adamantylzinc reagent. After screening several conditions and Mg sources, we found Mg with a particle size of −50 mesh at rt was optimal for reaction telescoping (Table 1, entry 4).¹³ One intriguing observation was that an activated Mg column could be stored under solvent (THF) for several weeks and reused for organozinc reagent formation without reactivating the column.

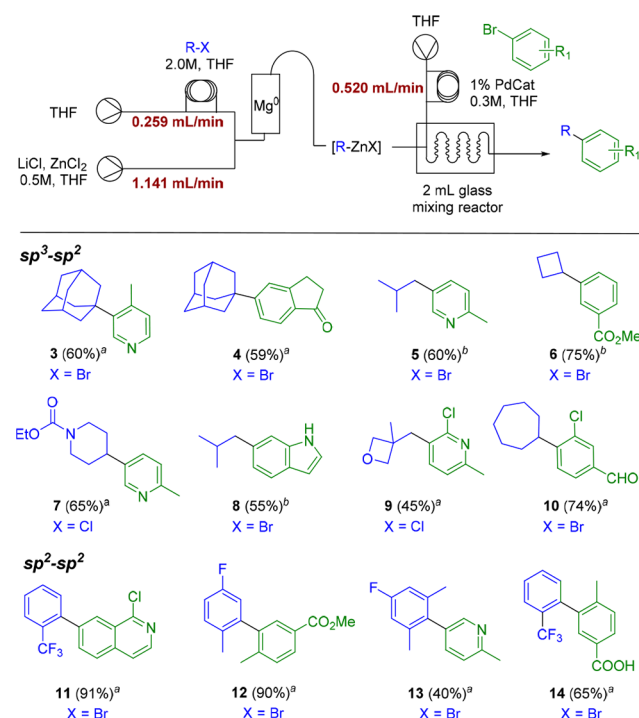
Generation of organozinc reagents was further simplified by pumping separate stock solutions of LiCl and ZnCl₂ with a solution of organohalide in flow via a T-mixer before passing through the Mg column with a residence time of 5 min (Table 2 flow scheme). This protocol allows rapid exchange of the organohalide and, when combined with an injection loop, can be used to generate organozinc reagents in a library format. An additional advantage of this flow protocol is the ability to simply inject organohalides to prepare the organozinc reagents on demand without reactivating the Mg column between each substrate. We next investigated a telescoping protocol where in situ generated organozinc reagents were subjected to Negishi cross-coupling reaction conditions with the goal of identifying an effective catalyst (Table 2). The in situ generated adamantylzinc reagent exiting the reactor was collected into a stirring THF solution of 3-bromo-4-methylpyridine while varying the Pd catalysts. Notable reactivity was observed with Pd(I) dimer¹⁵ and Pd(OAc)₂/Sphos.¹⁰ Reactions with both catalysts showed good conversion at rt within 5 min (Table 2, entries 1 and 2).

With the optimized set of flow and catalyst conditions in hand, we explored the substrate scope for Negishi cross-coupling reactions (Scheme 1).¹⁶ The in situ generated organozinc

Table 2. Negishi Coupling and Catalyst Optimization with Adamantylzinc Reagent


entry	catalyst	conversion (%) ^a
1	Pd(OAc) ₂ /Sphos	55
2	Pd(I) dimer	65
3	Pd-PEPPSI-IPr	20

^aReaction mixtures were stirred for 5 min at rt. Conversions were determined by LCMS analysis.

Scheme 1. Negishi (sp³–sp² and sp²–sp²) Cross-Coupling with In Situ Generated Organozinc Reagents[†]


[†]Reaction conditions: RX (3.3 equiv), LiCl (3.6 equiv), ZnCl₂ (3.6 equiv), ArBr (1.0 equiv), PdCat (0.01 equiv). ^aPalladium(I) dimer. ^bPd(OAc)₂/Sphos.

reagents were mixed downstream with a THF solution of aryl bromide and either Pd(I) dimer or Pd(OAc)₂/Sphos catalyst in a glass chip reactor at rt (Scheme 1) with a residence time of 1 min. A variety of substrates were amenable to organozinc formation with good conversion to the corresponding cross-coupling products with yields of 40–91%. Reactions with heterocycles bearing electron-donating groups (**3**, **5**, and **7**) or an electron-withdrawing group (**6**) proceeded efficiently with similarly good yields. A variety of functional groups on the aryl halide coupling partner, such as ketones, amines, aldehydes,

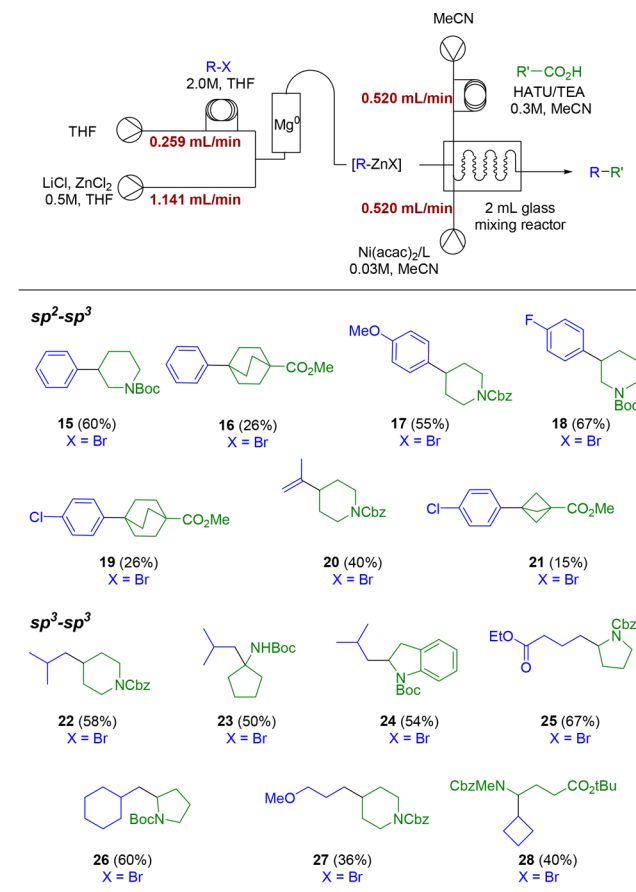
acids, and esters were well tolerated (4, 8, 10, 14, and 6, respectively) without hindrance of product formation. Furthermore, good chemoselectivity (Br vs Cl) was observed when using Pd(I) dimer (9, 10, and 11).¹⁵ A noteworthy feature of this flow protocol is its compatibility with both sp^3 – sp^2 and sp^2 – sp^2 carbon bond formation, which enabled formation of primary, secondary, and tertiary alkyl aryl cross-coupling products (5, 7, and 3, respectively). Additionally, sterically congested biaryl cross-coupling products (12, 13, and 14) form in modest to good yield.

We expanded the downstream protocols to include decarboxylative Negishi cross-coupling, reminiscent of recent work by the McMillan and Baran groups. McMillan demonstrated this transformation by directly combining carboxylic acids and organohalides under photoredox conditions.¹⁷ Although this method is attractive, implementation of this procedure requires specialized photoreactors and expensive Ir catalysts. Baran showed that organozinc reagents can be used in a nickel-mediated decarboxylative Negishi coupling with redox-active esters, that is, carboxylic acids activated with standard amide couplings reagents.^{4a,b,d,18} This transformation offers tremendous utility with access to C–C bond formations between sp^2 – sp^3 and sp^3 – sp^3 carbon centers from readily available carboxylic acids yet still requires inert and anhydrous atmosphere conditions to generate moisture-sensitive organozinc reagents, which can often be tedious in batch. This, combined with the stepwise synthesis of the redox-active esters, could limit its usage in an industrial medicinal chemistry setting.

Adapting our standardized in situ organozinc formation to decarboxylative Negishi coupling in flow required its own optimization. This work was initially focused on the sp^2 – sp^3 coupling of phenylzinc halide with 1-(*tert*-butoxycarbonyl)-3-piperidinecarboxylic acid to form compound 15 (Scheme 2) to probe other flow reaction parameters (i.e., nickel catalyst solubility and solvent compatibility). 15 was activated using HATU and triethylamine and then treated with different nickel catalysts in DMF and acetonitrile (see Supporting Information (SI)). Screening solvents and catalysts suggested that Ni(acac)₂/4,4'-dimethyl-2,2'-bipyridine in MeCN worked best in terms of solubility and conversion, while providing a productivity rate of approximately 5 mmol/h. Although other conditions (i.e., NiCl₂ in DMF) provided similar conversion, the catalyst was not practical to employ in flow due to poor solubility. Reactions in DMF also provided similar conversions; however, the solvent was not ideal due to formation of dimethyl amine amide byproducts and substrate precipitation in the reactor (see SI).

Under the optimized conditions, activation of the carboxylic acid in flow was performed by addition of HATU and triethylamine in acetonitrile. The resulting activated carboxylic acids were combined downstream with in situ generated organozinc reagents and catalyst and then passed through a glass reactor at rt with a residence time of 1 min. The scope of this flow protocol was explored and found to be remarkably broad (Scheme 2).¹⁶ Both electron-rich *para* methoxy and electron-poor *para* fluoro aromatic organozinc reagents were well tolerated, affording the corresponding arylated derivatives in respectable yields (17 and 18, respectively). Synthetically demanding biphenyl bioisosteres for medicinal chemistry were quickly generated by coupling readily available [2.2.2] and [1.1.1] bicyclo tertiary carboxylic acids, providing 16/19 and 21 in modest yields, respectively. Decarboxylative alkenylation^{18a} coupling was also achieved, rendering 20 in 40% yield.

Scheme 2. Decarboxylative Negishi (sp^2 – sp^3 and sp^3 – sp^3) Cross-Coupling with In Situ Generated Organozinc Reagents[†]



[†]Reaction conditions: RX (3.3 equiv), LiCl (3.6 equiv), ZnCl₂ (3.6 equiv), R'-CO₂H (1.0 equiv), HATU (1.0 equiv), TEA (1.0 equiv), Ni(acac)₂/L (0.1 equiv, 1:1.5). L = 4,4'-dimethyl-2,2'-bipyridine.

Decarboxylative Negishi cross-coupling in flow is applicable to both sp^2 – sp^3 and sp^3 – sp^3 couplings, requiring only a simple organohalide substrate change (Scheme 2). This further increases the value and utility of this method by enabling C–C bond formation between primary, secondary, and tertiary alkyl groups under flow conditions. For example, in situ generated alkyl organozinc reagents containing functional groups such as ester and methoxy undergo coupling with carboxylic acids to generate novel compounds 25 and 27, respectively. Additionally, sterically demanding amino acids can be modified efficiently using this protocol to form 28, suggesting the potential for late-stage modification or manufacturing of custom peptidomimetic derivatives. In all cases, coupling occurred smoothly and without interruption in the presence of protecting group functionality, i.e. esters and protected amines, which provides products suitable for further structural elaboration applicable to medicinal chemistry.

In summary, we have developed a versatile flow synthesis method for in situ formation of a wide variety of organozinc reagents prepared directly from readily available organohalides using a Mg column with direct transmetalation using ZnCl₂. Telescoping of the resulting organozinc reagents to Negishi (sp^3 – sp^2 and sp^2 – sp^2) and decarboxylative Negishi (sp^2 – sp^3 and sp^3 – sp^3) cross-coupling reactions avoids the need to isolate or transfer moisture-sensitive organozinc reagents. Modern flow

chemistry instrumentation features allow compatibility of this protocol to proceed in a single flow setup, in either one-off or library format. We anticipate that these advances will facilitate quick access to organozinc chemistry and cross-coupling adducts for medicinal chemists. Further exploration of the reactivity scope beyond this methodology and applications to the synthesis of complex molecules in drug discovery is in progress.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acs.orglett.8b03156](https://doi.org/10.1021/acs.orglett.8b03156).

Experimental procedures, characterization data, and copies of NMR spectra for all products (PDF)

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Notes

The authors declare no competing financial interest.

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